

Estimated Annual Agricultural Pesticide Use Washington State, Statewide Ranking 2014-2018

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Summary

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides **statewide in Washington** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data, summed across all Washington counties. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, each compound’s chemical class (abbreviated; see the key in Table 1 and the full classification in `data/pesticide_classification.csv`, shared with the Washington pesticide class trends chapter), and — for comparison — its 2018 rank within Yakima County alone. Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”); summing county estimates to a statewide total inherits this caveat.
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide (48 for Washington), versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019-included windows once a complete 2019 release is available.

A compound’s statewide rank and its Yakima County rank can differ substantially — a compound used heavily across WA’s wheat and row-crop counties but sparingly in Yakima’s orchards will rank higher statewide than in Yakima, and vice versa. NA in the Yakima rank column means the compound was not among Yakima County’s top 80 for 2018.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
ALS	ALS-Inhibitor (Sulfonylurea & Related)
AVM	Avermectin
BIO	Biological/Microbial
BIP	Bipyridylum
CB	Carbamate
CAA	Chloroacetanilide
DMI	DMI Fungicide (Triazole/Imidazole)
DIA	Diamide
DNA	Dinitroaniline
DTC	Dithiocarbamate

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
FUM	Fumigant
GLY	Glycine/Phosphonate
INO	Inorganic/Mineral
IGR	Insect Growth Regulator
NEO	Neonicotinoid
OC	Organochlorine
OP	Organophosphate
FUN	Other Fungicide
HRB	Other Herbicide
INS	Other Insecticide/Miticide
OTH	Other/Unclassified
PHX	Phenoxy
PHP	Phenylpyrazole
PGR	Plant Growth Regulator
PYR	Pyrethroid
SDH	SDHI
SPN	Spinosyn
QOI	Strobilurin (QoI)
TC	Thiocarbamate
TRZ	Triazine

Table 2: Top 80 EPest-high Estimates Statewide (WA), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) and Yakima County Rank

Rank (WA)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg	Rank (Yakima)
1	METAM	FUM	10983788	9543622	7815926	7
2	DICHLOROPROPENE	FUM	5403960	5738156	5790137	29
3	PETROLEUM OIL	INS	5294400	6161430	5527000	1
4	METAM POTASSIUM	FUM	2605463	1810088	1861955	13
5	SULFUR	INO	1790118	1484326	1383031	2
6	CHLOROPICRIN	FUM	1660175	655112	436589	28
7	GLYPHOSATE	GLY	1083658	1152865	1106914	6
8	BACILLUS AMYLOLIQUEFACIEN	BIO	1053895	707156	467499	92
9	CALCIUM POLYSULFIDE	INO	1047670	932494	816154	4
10	KAOLIN CLAY	INO	904020	1303433	973534	3
11	SULFURIC ACID	INO	673530	1436094	1476174	30
12	2,4-D	PHX	373773	312525	314410	18
13	COPPER	INO	318788	242440	192856	5
14	EPTC	TC	275642	249420	260421	64
15	PARAQUAT	BIP	240192	198784	162249	9
16	MANCOZEB	DTC	176597	230560	267681	21
17	CHLOROTHALONIL	FUN	132517	127428	121213	77
18	BROMOXYNIL	HRB	131784	146906	146611	75
19	MCPA	PHX	125980	131125	120874	89
20	CHLORPYRIFOS	OP	105926	126803	130985	8
21	GLUFOSINATE	GLY	104471	49569	38883	11
22	METOLACHLOR & METOLACHLOR-S	CAA	103224	93303	95188	16
23	METOLACHLOR-S	CAA	103224	93303	94395	17
24	PENDIMETHALIN	DNA	102888	117052	113673	23
25	BURKHOLDERIA SPP	BIO	85668	75310	56835	24
26	METRIBUZIN	TRZ	85508	82676	77113	40
27	OXAMYL	CB	73474	29107	52385	94
28	PROPICONAZOLE	DMI	73254	78181	62381	105
29	MCPB	PHX	72768	37024	19544	125
30	ATRAZINE	TRZ	69585	57434	44494	14

Table 2: Top 80 EPest-high Estimates Statewide (WA), 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) and Yakima County Rank (*continued*)

Rank (WA)	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg	Rank (Yakima)
31	DIMETHENAMID & DIMETHENAMID-P	CAA	66448	46147	39870	26
32	DIMETHENAMID-P	CAA	66448	46147	39870	27
33	COPPER HYDROXIDE	INO	65340	73064	65640	19
34	THIOPHANATE-METHYL	FUN	60584	77435	110334	35
35	BACILLUS THURINGIENSIS	BIO	57331	38416	29613	22
36	POTASSIUM BICARBONATE	INO	54119	65020	48582	12
37	CARBARYL	CB	51963	44469	62903	10
38	COPPER SULFATE	INO	49809	67180	60420	45
39	BENTAZONE	HRB	46270	46826	52941	120
40	DIMETHOATE	OP	37268	44926	47578	91
41	FLUROXYPYR	HRB	34132	26828	26127	140
42	COPPER OXYCHLORIDE	INO	33630	24397	16823	109
43	LINURON	HRB	33185	25863	18045	115
44	METHOMYL	CB	33086	45322	39755	112
45	DCPA	HRB	32618	56088	52597	155
46	DIQUAT	BIP	32613	20191	18750	107
47	ETHOPROPHOS	OP	31272	53990	40919	NA
48	DICAMBA	HRB	28797	36063	29664	73
49	CLOPYRALID	HRB	27852	19104	19125	98
50	TRIFLUMIZOLE	DMI	25283	27222	27074	15
51	BOSCALID	SDH	25236	31612	34289	42
52	PYRACLOSTROBIN	QOI	24775	29024	28932	34
53	MALEIC HYDRAZIDE	PGR	24123	26721	26700	118
54	FLUAZINAM	FUN	23658	17174	17242	116
55	ETHALFLURALIN	DNA	21573	23120	30586	184
56	FLUTOLANIL	SDH	21519	10580	11468	110
57	PENTHIOPYRAD	SDH	20843	17355	16278	49
58	SIMAZINE	TRZ	20248	9801	16724	32
59	TRIFLURALIN	DNA	20058	17725	13744	93
60	AZOXYSTROBIN	QOI	19527	27150	22575	104
61	DIURON	HRB	19142	42683	38795	44
62	MEFENOXAM	FUN	18496	15486	13455	81
63	PHOSPHORIC ACID	INO	17904	16564	49231	123
64	NORFLURAZON	HRB	16840	7399	10770	25
65	TRICLOPYR	HRB	16220	11995	10840	20
66	CLETHODIM	HRB	15959	11845	10565	127
67	TRI-ALLATE	TC	14878	48808	39868	177
68	BUPROFEZIN	IGR	14780	17663	14464	33
69	MALATHION	OP	13911	12943	18037	37
70	SPIROTETRAMAT	INS	12515	13025	11884	57
71	PROHEXADIONE	PGR	11986	6271	6661	31
72	FLUXAPYROXAD	SDH	11305	10717	7870	47
73	BIFENTHRIN	PYR	10626	13214	9578	83
74	CHLORANTRANILIPROLE	DIA	10418	9027	9413	36
75	SPIROMESIFEN	INS	10211	11246	10165	129
76	ACETAMIPRID	NEO	10145	7490	7104	38
77	AUREOBASIDIUM PULLULANS	BIO	10052	9243	5861	65
78	FLUOPYRAM	SDH	10034	9849	7725	41
79	FLUMIOXAZIN	HRB	9968	7344	4732	55
80	PROPARGITE	INS	9776	9617	13744	131

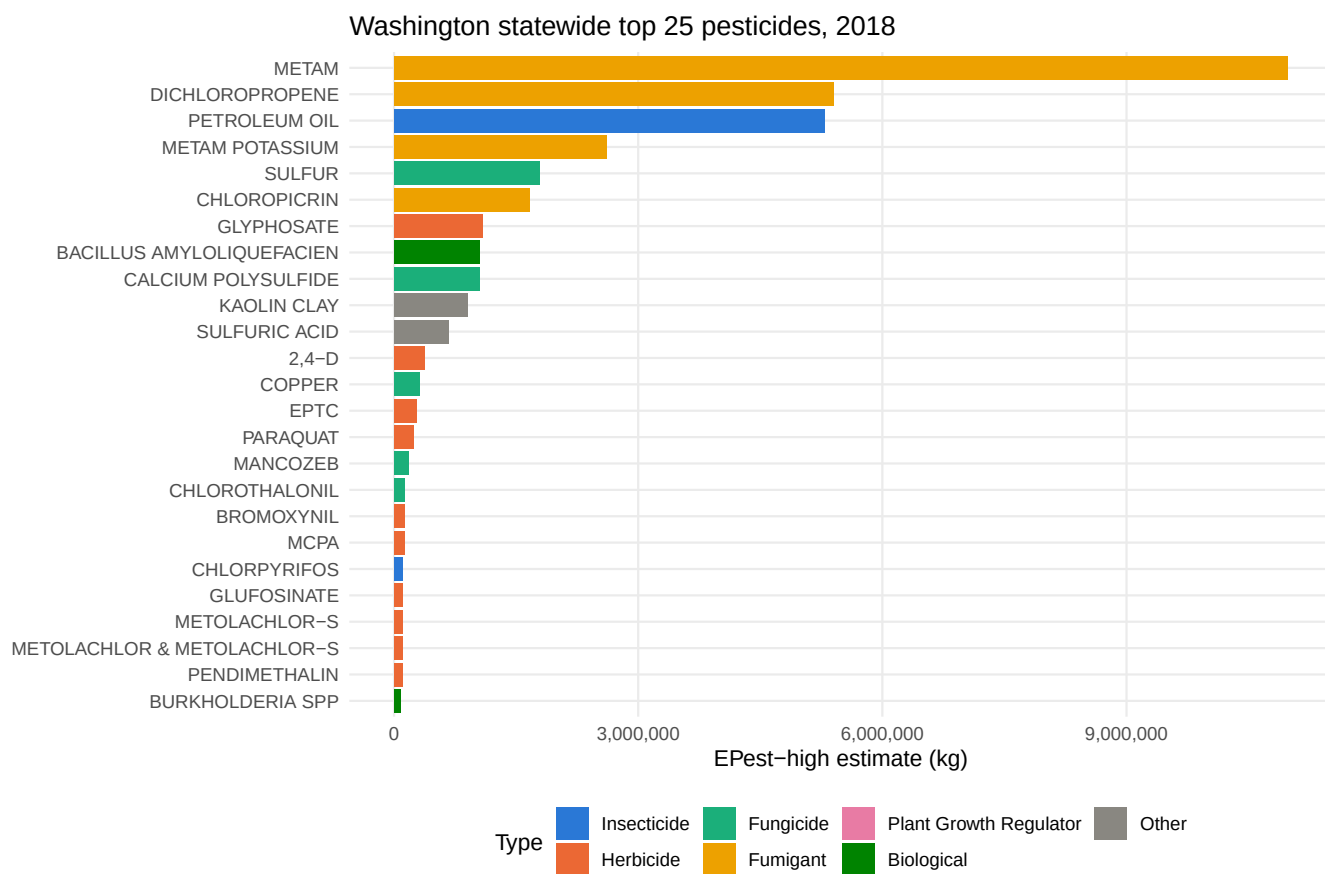


Figure 1: Top 25 compounds statewide by E Pest-high mass applied, most recent year, colored by pesticide type.

Appendices

Table 3: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	METAM	7959326	10983788
2	DICHLOROPROPENE	4950078	5403960
3	PETROLEUM OIL	5267665	5294400
4	METAM POTASSIUM	1782132	2605463
5	SULFUR	1770202	1790118
6	CHLOROPICRIN	336	1660175
7	GLYPHOSATE	994675	1083658
8	BACILLUS AMYLOLIQUEFACIEN	513383	1053895
9	CALCIUM POLYSULFIDE	1046460	1047670
10	KAOLIN CLAY	900921	904020
11	SULFURIC ACID	0	673530
12	2,4-D	323616	373773
13	COPPER	315290	318788
14	EPTC	155486	275642
15	PARAQUAT	209324	240192
16	MANCOZEB	167943	176597
17	CHLOROTHALONIL	90703	132517
18	BROMOXYNIL	117238	131784
19	MCPA	114100	125980
20	CHLORPYRIFOS	103286	105926
21	GLUFOSINATE	102997	104471
22	METOLACHLOR & METOLACHLOR-S	61167	103224
23	METOLACHLOR-S	61167	103224
24	PENDIMETHALIN	84384	102888
25	BURKHOLDERIA SPP	42896	85668
26	METRIBUZIN	49710	85508
27	OXAMYL	54503	73474
28	PROPICONAZOLE	66627	73254
29	MCPB	7431	72768
30	ATRAZINE	53269	69585
31	DIMETHENAMID & DIMETHENAMID-P	46567	66448
32	DIMETHENAMID-P	46567	66448
33	COPPER HYDROXIDE	44224	65340
34	THIOPHANATE-METHYL	53986	60584
35	BACILLUS THURINGIENSIS	57235	57331
36	POTASSIUM BICARBONATE	34901	54119
37	CARBARYL	50998	51963
38	COPPER SULFATE	48332	49809
39	BENTAZONE	40198	46270
40	DIMETHOATE	27360	37268
41	FLUROXYPYR	33425	34132
42	COPPER OXYCHLORIDE	23454	33630
43	LINURON	24301	33185
44	METHOMYL	24892	33086
45	DCPA	30980	32618
46	DIQUAT	22664	32613
47	ETHOPROPHOS	322	31272
48	DICAMBA	23019	28797
49	CLOPYRALID	16755	27852
50	TRIFLUMIZOLE	25156	25283

Table 4: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	8318982	9543622
2	PETROLEUM OIL	5952023	6161430
3	DICHLOROPROPENE	5286516	5738156
4	METAM POTASSIUM	1162019	1810088
5	SULFUR	1445992	1484326
6	SULFURIC ACID	0	1436094
7	KAOLIN CLAY	1066988	1303433
8	GLYPHOSATE	1088891	1152865
9	CALCIUM POLYSULFIDE	903338	932494
10	BACILLUS AMYLOLIQUEFACIEN	401943	707156
11	CHLOROPICRIN	69590	655112
12	2,4-D	276231	312525
13	EPTC	151354	249420
14	COPPER	238436	242440
15	MANCOZEB	226021	230560
16	PARAQUAT	157527	198784
17	BROMOXYNIL	127219	146906
18	MCPA	109341	131125
19	CHLOROTHALONIL	110353	127428
20	CHLORPYRIFOS	108461	126803
21	PENDIMETHALIN	94092	117052
22	METOLACHLOR & METOLACHLOR-S	44851	93303
23	METOLACHLOR-S	44851	93303
24	METRIBUZIN	60225	82676
25	PROPICONAZOLE	71533	78181
26	THIOPHANATE-METHYL	62350	77435
27	BURKHOLDERIA SPP	36992	75310
28	COPPER HYDROXIDE	56133	73064
29	COPPER SULFATE	60603	67180
30	POTASSIUM BICARBONATE	48555	65020
31	ATRAZINE	39906	57434
32	DCPA	54596	56088
33	PROPAMOCARB HCL	25475	54430
34	ETHOPROPHOS	23113	53990
35	GLUFOSINATE	42712	49569
36	TRI-ALLATE	11284	48808
37	BENTAZONE	43616	46826
38	DIMETHENAMID & DIMETHENAMID-P	32533	46147
39	DIMETHENAMID-P	32533	46147
40	METHOMYL	41036	45322
41	DIMETHOATE	28678	44926
42	CARBARYL	42621	44469
43	DIURON	9352	42683
44	BACILLUS THURINGIENSIS	35267	38416
45	ZIRAM	12839	37829
46	MCPB	3932	37024
47	DICAMBA	22533	36063
48	ZINC	26512	32277
49	BOSCALID	28121	31612
50	OXAMYL	20604	29107

Table 5: Top 50 Statewide Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	6738971	7815926
2	DICHLOROPROPENE	5298824	5790137
3	PETROLEUM OIL	5397589	5527000
4	METAM POTASSIUM	1275353	1861955
5	SULFURIC ACID	0	1476174
6	SULFUR	1340868	1383031
7	GLYPHOSATE	1057130	1106914
8	KAOLIN CLAY	828825	973534
9	CALCIUM POLYSULFIDE	795459	816154
10	BACILLUS AMYLOLIQUEFACIEN	268663	467499
11	CHLOROPICRIN	59546	436589
12	2,4-D	272191	314410
13	MANCOZEB	261335	267681
14	EPTC	174025	260421
15	COPPER	188974	192856
16	PARAQUAT	126074	162249
17	BROMOXYNIL	125210	146611
18	CHLORPYRIFOS	114945	130985
19	CHLOROTHALONIL	110872	121213
20	MCPA	105326	120874
21	PENDIMETHALIN	95587	113673
22	THIOPHANATE-METHYL	76257	110334
23	METOLACHLOR & METOLACHLOR-S	48597	95188
24	METOLACHLOR-S	48184	94395
25	METRIBUZIN	57389	77113
26	COPPER HYDROXIDE	53936	65640
27	CARBARYL	61457	62903
28	PROPICONAZOLE	57444	62381
29	PINOLENE	29216	60585
30	COPPER SULFATE	49264	60420
31	BURKHOLDERIA SPP	27955	56835
32	PROPAMOCARB HCL	25475	54430
33	BENTAZONE	48461	52941
34	DCPA	46905	52597
35	OXAMYL	45786	52385
36	PHOSPHORIC ACID	5371	49231
37	POTASSIUM BICARBONATE	38444	48582
38	DIMETHOATE	31300	47578
39	CAPTAN	8495	44992
40	ATRAZINE	32726	44494
41	ETHOPROPHOS	18699	40919
42	DIMETHENAMID & DIMETHENAMID-P	27054	39870
43	DIMETHENAMID-P	27054	39870
44	TRI-ALLATE	12790	39868
45	METHOMYL	35129	39755
46	GLUFOSINATE	28205	38883
47	DIURON	11816	38795
48	ZIRAM	20984	36024
49	BOSCALID	30227	34289
50	CUPROUS OXIDE	19282	32611